



FACULTY OF ACCOUNTING AND INFORMATICS

DEPARTMENT OF INFORMATION TECHNOLOGY

INTEGRATIVE PROGRAMMING AND TECHNOLOGIES III

TEST 2

DATE: 4 MAY 2026

TIME: 45 MINS

EXAMINER: BOAMAH-ABU, CHARLES (PhD)

TOTAL MARKS: 36

NO. OF PAGES: 3 (Including cover page)

INSTRUCTIONS/REQUIREMENTS

1. This examination is closed book
2. Answer all questions
3. Where stated "Briefly answer", please give a short answer
4. Where stated "Write the missing parts" of code, please write only the missing parts
5. Ensure your handwriting is readable

Question 1

10 marks

1. True or false: XML stores data as text. (1)
2. True or false: XML allows you to create your own tags to store data. (1)
3. The table below shows XML technologies and their uses. Match the technology to the appropriate use (Write **only** the **number** of a technology and the **letter** corresponding to the use). (4)

	<i>Tools</i>		<i>Used to</i>
1	XSLT	A	Define the structure of XML data
2	XML (Language)	B	Write the XML code to store data
3	XPath	C	Transform how XML data must be displayed
4	XML Schema	D	Navigate to XML nodes or values

4. Briefly explain why is XML is important in Integrative programming? (2)
5. What is the main advantage of Stax over DOM? (2)

Question 2

26

The table below shows part of the data on lecturers in the IT Department. Each lecturer has **id**, **name**, **surname** and **phone** number.

1	Sharon	Amp	0612122231
2	Bertha	Boa	0612122232
3	Ngcwele	Mat	0612122233

The data is to be stored as XML in a file called **lecturers.xml**. The root element is **Root**. This contains several **Lecturer** elements. Each **Lecturer** element consists of **id** (an attribute, required) and two elements: **fullname** and **phone**. The **fullname** element consists of **name** and **surname** elements.

1. Complete the XML code (below) for the **first** lecturer in **lecturer.xml** (use **id** as attribute and **Lecturer**, **fullname**, **name**, **surname** and **phone** as tags). (5)

```
<?xml version="1.0" encoding="UTF-8"?>
<Root>
  _____
  _____
  _____
  _____
  _____
  _____
  ...
</Root>
```

2. Write the missing parts of the DTD code in a file **lecturer.dtd**, which defines the structure of the XML in **lecturer.xml**. (9)

```
<?xml version="1.0" encoding="UTF-8"?>
<!ELEMENT Root (Lecturer)+>
<!-->
<!-->
<!-->
<!-->
<!-->
<!-->
```

3. Link the XML code in **lecturer.xml** to the DTD in **lecturer.dtd**. (2)

```
<!DOCTYPE _____ SYSTEM "_____">
```

4. XSL code in a file **lecturer.xsl** defines how the XML in **lecturer.xml** is to be displayed in a web page as shown below.

Id	Name	Surname	Phone
1	Sharon	Amp	0612122231
2	Bertha	Boa	0612122232
3	Ngcwele	Mat	0612122233

Write the missing parts of the XSL code (below) in **lecturer.xsl**, which extracts the data in the XML in **lecturer.xml**. (8)

```
<xsl:template match="/">
...
...
  <xsl:for-each select="_____">
    <xsl:_____ = _____>
    <xsl:_____ = _____>
    <xsl:_____ = _____>
    <xsl:_____ = _____>
  </xsl:for-each>
...
...
</xsl:template>
```

5. Write the missing parts of the line of code (below) in **lecturer.xml**, which links the XML to the XSL in **lecturer.xsl**. (2)

```
<?xml-stylesheet type="_____" href="_____"?>
```